Supplementary File 1. PRISMA checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4-5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4-5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5, Supplemnetary File
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5-6, Supplemnetary File
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5-6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6-7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Page 6, Figure 2.
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Page 6, Figure 2
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7-8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8-9, Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8-9. Table 1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Page 8, Figure 2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9-11, Figure 3
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	9-11, Figure 3
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Figure 2
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10-11, Figure 4
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	11-12
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	12-15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15-16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	8

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: **www.prisma-statement.org**.

Supplementary File 2. Database searches

Database	Date range searched	Date searched	Number of results		
			GLP-1	DPP-4	SGLT-2i
Cochrane Central Register of Controlled Trials	1946 - current	13/02/20	1102	830	743
Ovid–EMBASE	1974 – February 13th 2021	13/02/20	3231	3791	3351
Ovid–HMIC (Health Management Information Consortium)	1979 to November 2020	13/02/20	0	0	0
Ovid–MEDLINE	1946 - current	13/02/20	483	472	450
Ovid–MEDLINE E-pub ahead of print	February 13th 2021	13/02/20	6	2	12
Ovid–MEDLINE In-Process and Other Non-Indexed Citations	1946 - current	13/02/20	465	470	434
PubMed	1963 - current	13/02/20	333	210	178
Scopus	-	13/02/20	6	1	1

Supplementary File 3. Funnel plot of publication bias

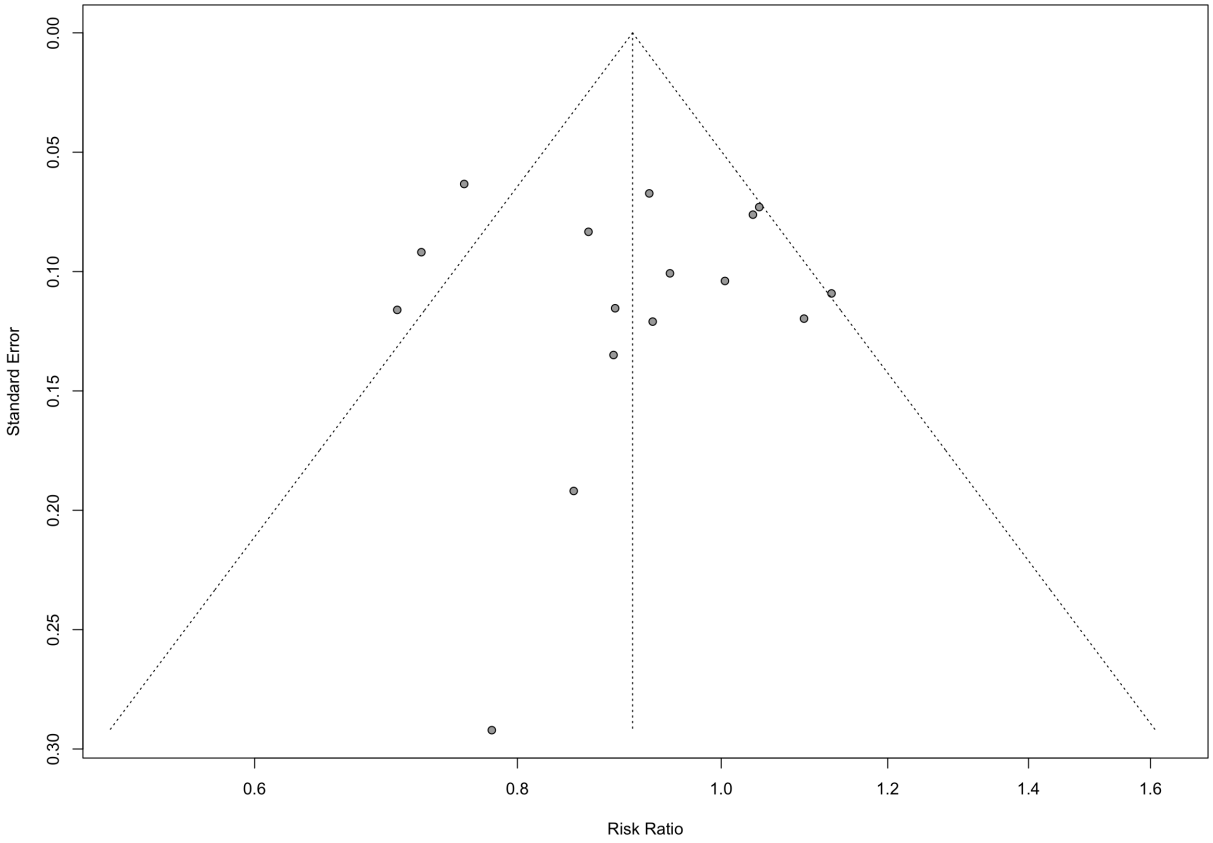
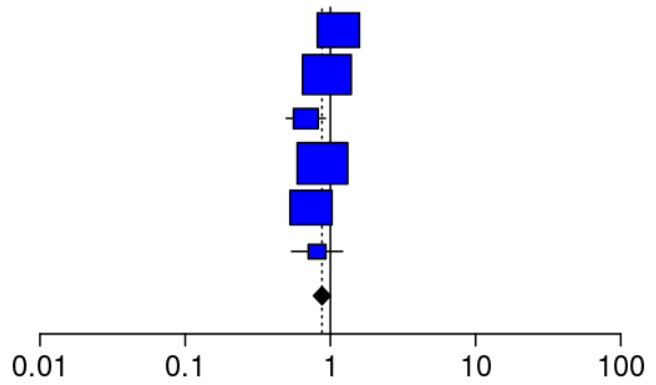


Figure 4a. Fragility Index in GLP-1 analogue study participants

eGFR ≥60ml/min

Study	GLP-1 analogue Arm A Events	Totals	Placebo Arm B Events	Totals	Weight	Risk Ratio MH, Random, 95% CI
ELIXA	268	2370	**227**	2278	18.6%	1.13 [0.96--1.34]
LEADER	436	3552	471	3630	21.6%	0.95 [0.84--1.07]
SUSTAIN-6	63	1179	93	1179	10.9%	0.68 [0.50--0.92]
EXSCEL	549	5769	620	5745	22.4%	0.88 [0.79--0.98]
HARMONY	222	3633	300	3608	18.6%	0.73 [0.62--0.87]
PIONEER-6	41	1150	51	1158	7.8%	0.81 [0.54--1.21]
Total (95%CI)		**17653**		**17598**		**0.88 [0.77--1.00]**

Heterogeneity: I2 = 71%

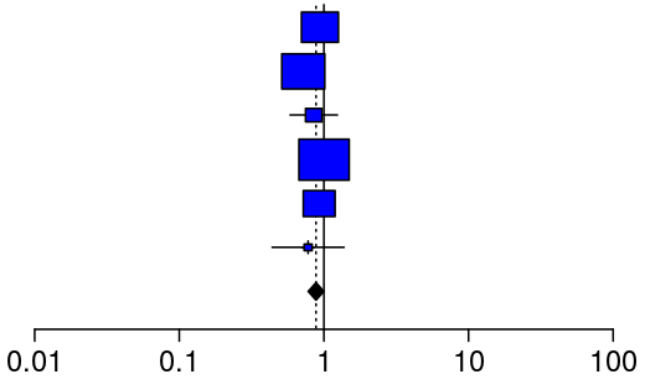


After 14 specific event-status modifications, the conclusion of the meta-analysis was turned to statistically non-significant (Fragility index = 14)

eGFR <60ml/min

Study	GLP-1 analogue Arm A Events	Totals	Placebo Arm B Events	Totals	Weight	Risk Ratio MH, Random, 95% CI
ELIXA	**129**	659	156	748	19.8%	0.94 [0.76--1.16]
LEADER	172	1116	223	1042	23.1%	0.72 [0.60--0.86]
SUSTAIN-6	45	469	53	470	8.8%	0.85 [0.58--1.24]
EXSCEL	**274**	1565	284	1626	27.0%	1.00 [0.86--1.17]
HARMONY	116	1098	128	1124	17.1%	0.93 [0.73--1.18]
PIONEER-6	20	434	25	422	4.3%	0.78 [0.44--1.38]
Total (95%CI)		**5341**		**5432**		**0.88 [0.78--1.00]**

Heterogeneity: I2 = 40%



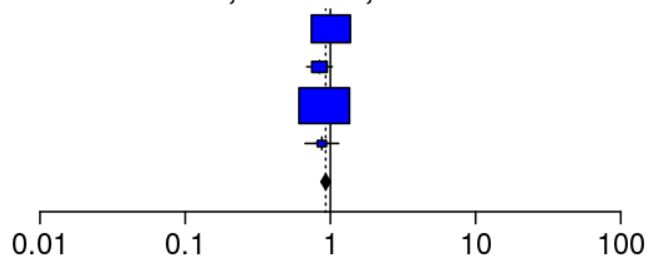
After 18 specific event-status modifications, the conclusion of the meta-analysis was turned to statistically significant (Fragility index = 18)

Figure 4b. Fragility Index in DPP-4 inhibitor study participants

eGFR ≥60ml/min

Study	DPP-4 inhibitor Arm A Events	DPP-4 inhibitor Arm A Totals	Placebo Arm B Events	Placebo Arm B Totals	Weight	Risk Ratio MH, Random, 95% CI
SAVOR-TIMI 53	**451**	6986	**444**	6930	34.3%	1.01 [0.89--1.14]
EXAMINE	160	1929	**186**	1886	13.6%	0.84 [0.69--1.03]
TECOS	**539**	6646	**595**	6656	44.3%	0.91 [0.81--1.01]
CARMELINA	**93**	1294	110	1337	7.8%	0.87 [0.67--1.14]
Total (95%CI)		**16855**		**16809**		**0.93 [0.86--1.00]**

Heterogeneity: I2 = 0%

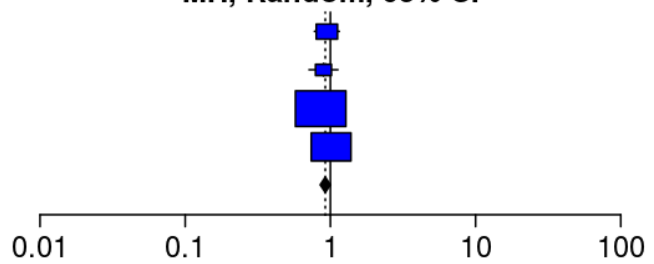


After 89 specific event-status modifications, the conclusion of the meta-analysis was turned to statistically significant (Fragility index = 89)

eGFR <60ml/min

Study	DPP-4 inhibitor Arm A Events	DPP-4 inhibitor Arm A Totals	Placebo Arm B Events	Placebo Arm B Totals	Weight	Risk Ratio MH, Random, 95% CI
SAVOR-TIMI 53	167	1294	175	1282	16.7%	0.95 [0.78--1.15]
EXAMINE	**116**	772	**133**	793	12.5%	0.90 [0.71--1.13]
TECOS	259	686	**301**	683	39.6%	0.86 [0.75--0.97]
CARMELINA	**321**	2200	310	2148	31.2%	1.01 [0.88--1.17]
Total (95%CI)		**4952**		**4906**		**0.92 [0.85--1.00]**

Heterogeneity: I2 = 1%



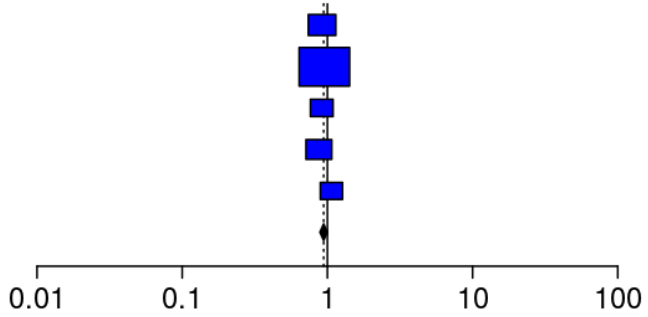
After 62 specific event-status modifications, the conclusion of the meta-analysis was turned to statistically significant (Fragility index = 62)

Figure 4c. Fragility Index in SGLT-2 inhibitor study participants

eGFR ≥60ml/min

Study	SGLT-2 inhibitor Arm A Events	Totals	Placebo Arm B Events	Totals	Weight	Risk Ratio MH, Random, 95% CI
CANVAS	418	4684	332	3417	18.5%	0.92 [0.80--1.05]
DECLARE-TIMI-58	671	7975	699	7919	34.0%	0.95 [0.86--1.05]
EMPA-REG	416	3475	**226**	1726	15.2%	0.91 [0.79--1.06]
VERTIS Trial	465	4295	**265**	2138	17.4%	0.87 [0.76--1.01]
EMPEROR-Reduced	**255**	969	**237**	960	14.9%	1.07 [0.92--1.24]
Total (95%CI)		**21398**		**16160**		**0.94 [0.89--1.00]**

Heterogeneity: I2 = 0%

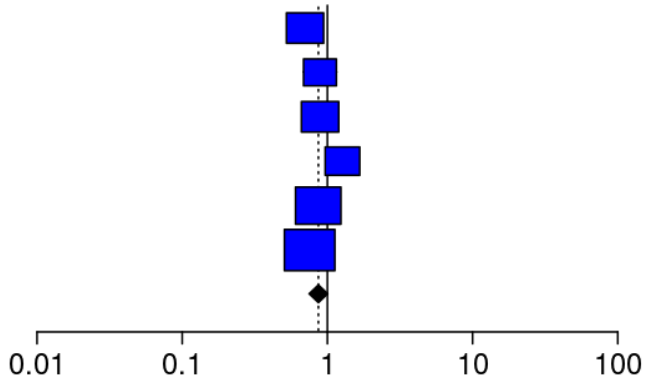


After 85 specific event-status modifications, the conclusion of the meta-analysis was turned to statistically significant (Fragility index = 85)

eGFR <60ml/min

Study	SGLT-2 inhibitor Arm A Events	Totals	Placebo Arm B Events	Totals	Weight	Risk Ratio MH, Random, 95% CI
CANVAS	119	1110	142	929	15.6%	0.70 [0.56--0.88]
DECLARE-TIMI-58	85	606	104	659	13.8%	0.89 [0.68--1.16]
EMPA-REG	176	1212	99	607	15.7%	0.89 [0.71--1.12]
VERTIS Trial	188	1198	**75**	607	14.5%	1.27 [0.99--1.63]
EMPEROR-Reduced	202	893	237	906	19.1%	0.86 [0.73--1.02]
SCORED	400	5292	530	5292	21.2%	0.75 [0.67--0.85]
Total (95%CI)		**10311**		**9000**		**0.87 [0.75--1.00]**

Heterogeneity: I2 = 69%



After 12 specific event-status modification, the conclusion of the meta-analysis was turned to statistically non-significant (Fragility index = 12)